

SECTION AAnswer **all** questions.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in sodium oxide. [2]

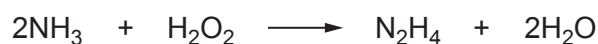
2. Complete the following sentence: [1]

Hydrogen bonding is an intermolecular force that occurs between molecules containing hydrogen atoms bonded to small, atoms which have lone pairs of electrons for example

3. By inserting arrows to represent electrons, show the electronic configuration of a calcium atom. [1]

1s	2s	2p	3s	3p	3d	4s

4. Hydrazine can be manufactured from ammonia.



Calculate the atom economy of this reaction. [1]

atom economy = %

